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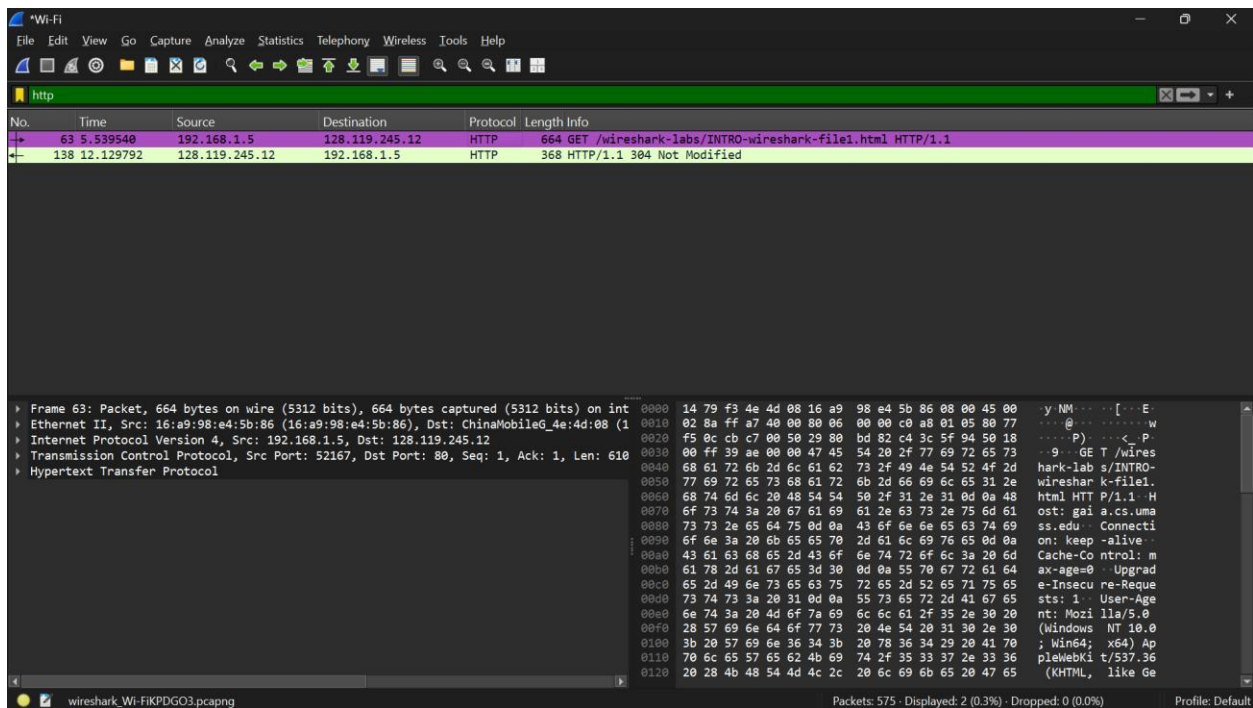
Roll # BSDSF24A050

Assignment

Wireshark intro

Assignment No. 1

This is the main page of the wireshark for the packet record



This is the GET Request page for (gaia.cs.umass.edu/wireshark-labs/INTRO-wireshark-file1.html)

The image shows a Wireshark packet capture window. The top pane displays the packet list, with packet 63 selected. The middle pane shows the packet details for the selected packet, which is an HTTP GET request. The bottom pane shows the raw packet bytes in hexadecimal and ASCII.

Packet List:

- Frame 63: Packet, 664 bytes on wire (5312 bits), 664 bytes captured (5312 bits) on interface \Device\NPF_{B5A8EC7D-CC2C-4702-B36C-D99E6968836A}, id 0
- Ethernet II, Src: 16:a9:98:e4:5b:86 (16:a9:98:e4:5b:86), Dst: ChinaMobileG_4e:4d:08 (14:79:f3:4e:4d:08)
- Internet Protocol Version 4, Src: 192.168.1.5, Dst: 128.119.245.12
- Transmission Control Protocol, Src Port: 52167, Dst Port: 80, Seq: 1, Ack: 1, Len: 610
- Hypertext Transfer Protocol
- GET /wireshark-labs/INTRO-wireshark-file1.html HTTP/1.1

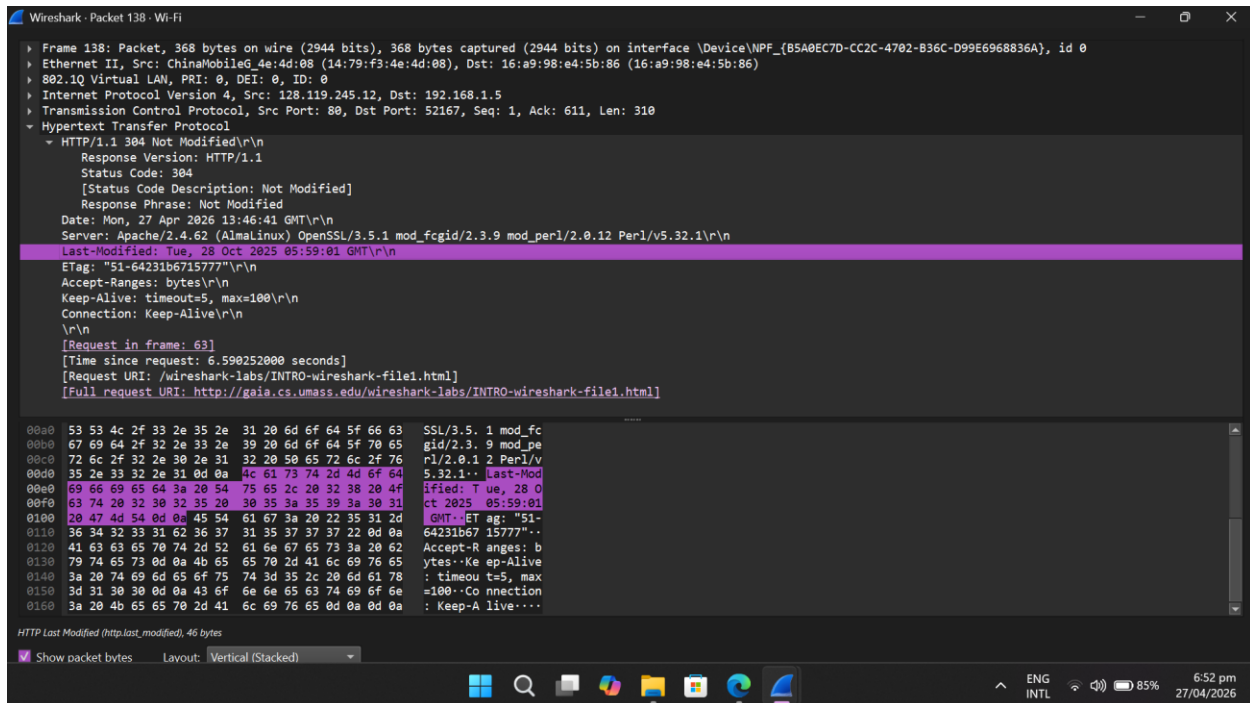
Packet Details:

- Request Method: GET
- Request URI: /wireshark-labs/INTRO-wireshark-file1.html
- Request Version: HTTP/1.1
- Host: gaia.cs.umass.edu
- Connection: keep-alive
- Cache-Control: max-age=0
- Upgrade-Insecure-Requests: 1
- User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36 Edg/147.0.0.0
- Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7
- Accept-Encoding: gzip, deflate
- Accept-Language: en-US,en;q=0.9
- If-None-Match: "51-64231b6715777"
- If-Modified-Since: Tue, 28 Oct 2025 05:59:01 GMT

Raw Packet Bytes:

```
0030 00 ff 39 ae 00 00 47 45 54 20 2f 77 69 72 65 73 09 20 2f 77 69 72 65 73
0040 68 61 72 6b 2d 6c 61 62 73 2f 49 4e 54 52 4f 2d 68 61 72 6b 2d 66 69 6c 65 31 2e
0050 77 69 72 65 73 68 61 72 6b 2d 66 69 6c 65 31 2e 68 74 6d 6c 20 48 54 54 50 2f 31 2e 31 0d 0a 48
0060 6f 73 74 3a 20 67 61 69 61 2e 63 73 2e 75 6d 61 73 73 2e 65 64 75 0d 0a 43 6f 6e 65 63 74 69
0070 6f 6e 3a 20 6b 65 65 70 2d 61 6c 69 76 65 0d 0a 43 61 63 68 65 2d 43 6f 6e 74 72 6f 6c 3a 20 6d
0080 61 78 2d 61 67 65 3d 30 0d 0a 55 70 67 72 61 64 65 2d 49 6e 73 65 63 75 72 65 2d 52 65 71 75 65
0090 73 74 73 3a 20 31 0d 0a 55 73 65 72 2d 41 67 65 6e 74 3a 20 4d 6f 7a 69 6c 6c 61 2f 35 2e 30 20
00a0 28 57 69 6e 64 6f 77 73 20 4e 54 20 31 30 2e 30 3b 20 57 69 6e 36 34 3b 20 78 36 34 29 20 41 70
```

This is the Response page.



Question Answers:

1. Protocol = TCP, HTTP, DNS
2. Time = 6.59 seconds
3. IP address :
 - Client (my computer): **192.168.1.5**
 - Server (gaia.cs.umass.edu): **128.119.245.12**
4. User-Agent field (browser type):
Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36 Edg/147.0.0.0 (Microsoft edge browser).
5. Destination Port in TCP details:
Destination port: **80** (standard HTTP port).

6. Printing GET and OK packets:

C:\Users\umaru\AppData\Local\Temp\wireshark_Wi-FiKPDGO3.pcapng 575 total packets, 2 shown No. Time Source Destination Protocol Length Info 63 2026-04-27 18:46:33.429299 192.168.1.5 128.119.245.12 HTTP 664 GET /wireshark-labs/INTRO-wireshark-file1.html HTTP/1.1 Frame 63: Packet, 664 bytes on wire (5312 bits), 664 bytes captured (5312 bits) on interface \Device\NPF_{B5A0EC7D-CC2C-4702-B36C-D99E6968836A}, id 0 Ethernet II, Src: 16:a9:98:e4:5b:86 (16:a9:98:e4:5b:86), Dst: ChinaMobileG_4e:4d:08 (14:79:f3:4e:4d:08) Internet Protocol Version 4, Src: 192.168.1.5, Dst: 128.119.245.12 Transmission Control Protocol, Src Port: 52167, Dst Port: 80, Seq: 1, Ack: 1, Len: 610 Hypertext Transfer Protocol GET /wireshark-labs/INTRO-wireshark-file1.html HTTP/1.1\r\n Request Method: GET Request URI: /wireshark-labs/INTRO-wireshark-file1.html Request Version: HTTP/1.1 Host: gaia.cs.umass.edu\r\n Connection: keep-alive\r\n Cache-Control: max-age=0\r\n Upgrade-Insecure-Requests: 1\r\n User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36 Edg/147.0.0.0\r\n Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7\r\n Accept-Encoding: gzip, deflate\r\n Accept-Language: en-GB,en;q=0.9,en-US;q=0.8\r\n If-None-Match: "51-64231b6715777"\r\n If-Modified-Since: Tue, 28 Oct 2025 05:59:01 GMT\r\n \r\n [Response in frame: 138] [Full request URI: http://gaia.cs.umass.edu/wireshark-labs/INTRO-wireshark-file1.html] No. Time Source Destination Protocol Length Info 138 2026-04-27 18:46:40.019551 128.119.245.12 192.168.1.5 HTTP 368 HTTP/1.1 304 Not Modified Frame 138: Packet, 368 bytes on wire (2944 bits), 368 bytes captured (2944 bits) on interface \Device\NPF_{B5A0EC7D-CC2C-4702-B36C-D99E6968836A}, id 0 Ethernet II, Src: ChinaMobileG_4e:4d:08 (14:79:f3:4e:4d:08), Dst: 16:a9:98:e4:5b:86 (16:a9:98:e4:5b:86) 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 0 Internet Protocol Version 4, Src: 128.119.245.12, Dst: 192.168.1.5 Transmission Control Protocol, Src Port: 80, Dst Port: 52167, Seq: 1, Ack: 611, Len: 310 Hypertext Transfer Protocol HTTP/1.1 304 Not Modified\r\n Response Version: HTTP/1.1 Status Code: 304 [Status Code Description: Not Modified] Response Phrase: Not Modified Date: Mon, 27 Apr 2026 13:46:41 GMT\r\n Server: Apache/2.4.62 (AlmaLinux) OpenSSL/3.5.1 mod_fcgid/2.3.9 mod_perl/2.0.12 Perl/v5.32.1\r\n Last-Modified: Tue, 28 Oct 2025 05:59:01 GMT\r\n ETag: "51-64231b6715777"\r\n Accept-Ranges: bytes\r\n Keep-Alive: timeout=5, max=100\r\n Connection: Keep-Alive\r\n \r\n [Request in frame: 63] [Time since request: 6.590252000 seconds] [Request URI: /wireshark-labs/INTRO-wireshark-file1.html] [Full request URI: <http://gaia.cs.umass.edu/wireshark-labs/INTRO-wireshark-file1.html>]

Wireshark HTTP

Assignment No. 2

1. The Basic HTTP GET/response interaction

Q1: Browser & Server HTTP Version

- Browser request line: GET ... HTTP/1.1 → Browser is using HTTP/1.1
- Server response line: HTTP/1.1 304 Not Modified → Server is also using HTTP/1.1

Q2: Accepted Languages:

Accept-Language: en-GB,en;q=0.9,en-US;q=0.8

Browser indicates it can accept British English, generic English, and US English.

Q3: IP Addresses

- Client (your computer): 192.168.1.5
- Server (gaia.cs.umass.edu): 128.119.245.12

Q4: Status Code

- Response line: HTTP/1.1 304 Not Modified → Status code = **304 (Not Modified)**

Q5: Last Modified Date From the response headers:

- Last-Modified: Tue, 28 Oct 2025 05:59:01 GMT → File was last modified on **28 October 2025, 05:59:01 GMT**

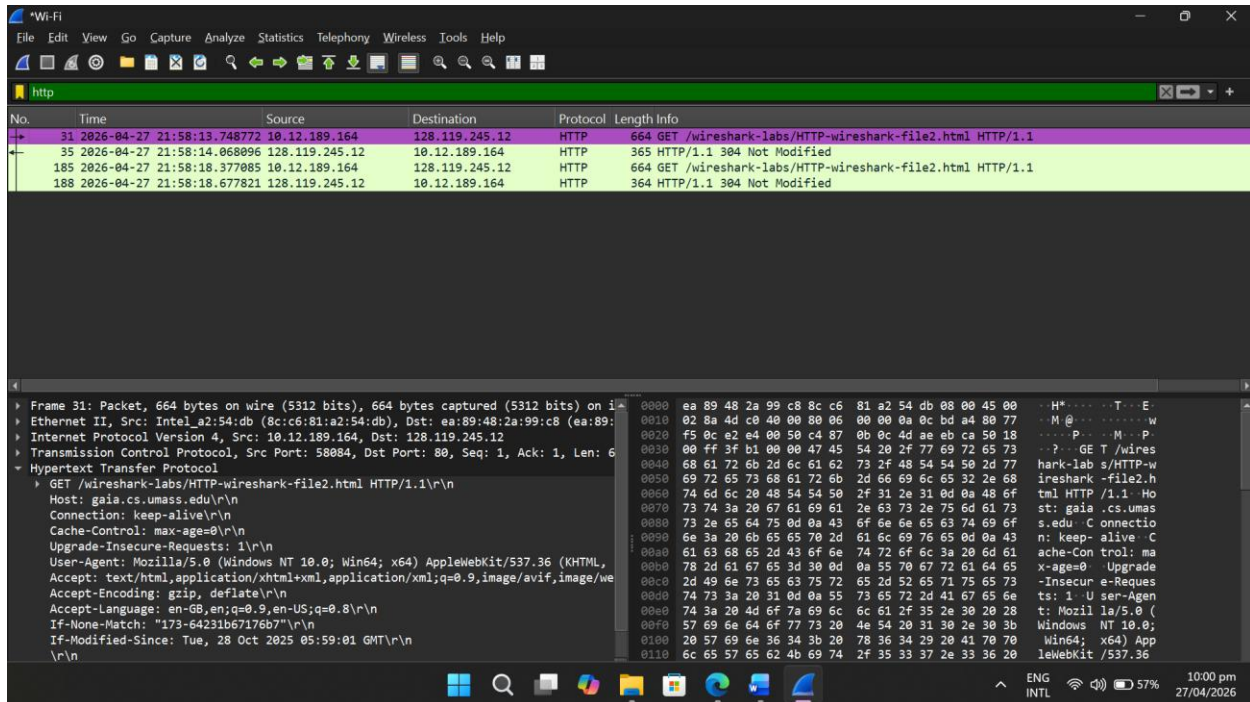
Q6: Content Length

- Response headers do **not include Content-Length** (because 304 Not Modified responses don't return a body). → **0 bytes of content returned** in this case.

Q7: Hidden Headers in Raw Data

Example hidden header: If-None-Match

2. The HTTP CONDITIONAL GET/response interaction



Q8. First HTTP GET request — “If-Modified-Since” line?

- **No.** The very first GET request does **not** contain an If-Modified-Since header.

Q9. First server response — did the server return the file contents? How can you tell?

- **Yes.** The server responded with **200 OK** and returned the full HTML file.

Q10. Second HTTP GET request — “If-Modified-Since” line? What follows?

- **Yes.** The second GET request includes:

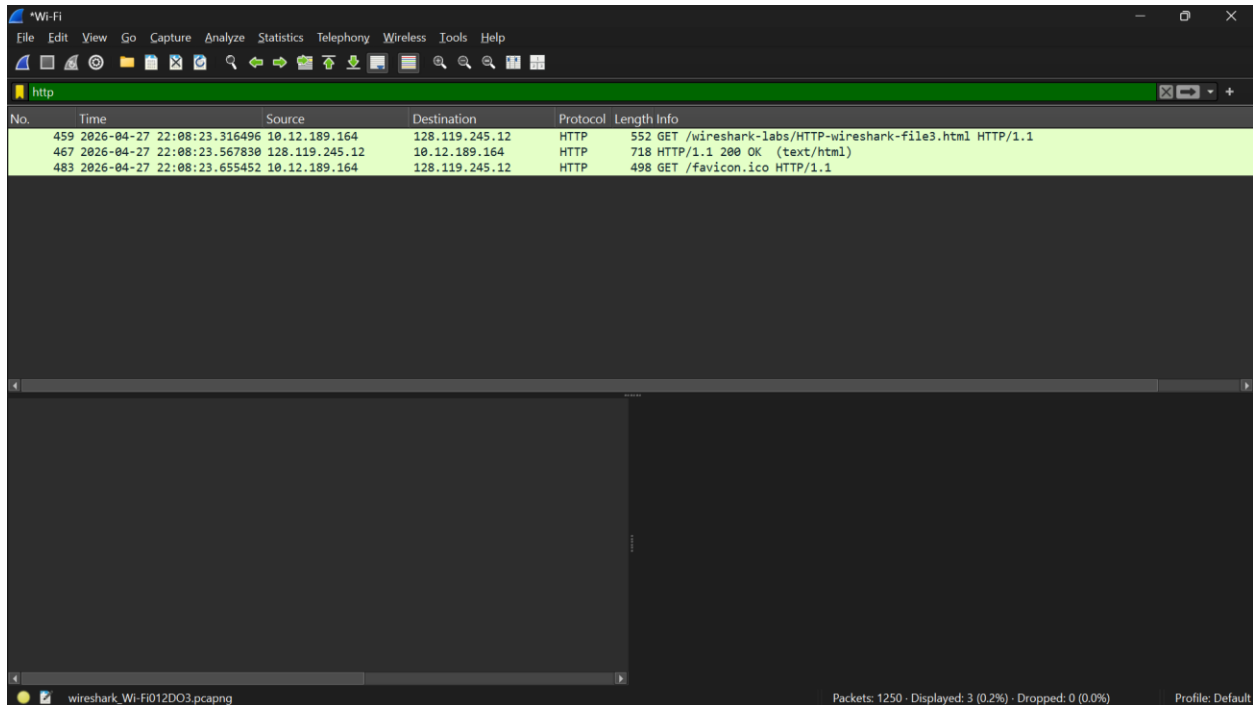
Code

If-Modified-Since: Tue, 28 Oct 2025 05:59:01 GMT

Q11. Second server response — status code and phrase? Did the server return the file?

- Status code: **304 Not Modified**

3. Retrieving Long Documents



No.	Time	Source	Destination	Protocol	Length	Info
459	2026-04-27 22:08:23.316496	10.12.189.164	128.119.245.12	HTTP	552	GET /wireshark-labs/HTTP-wireshark-file3.html HTTP/1.1
467	2026-04-27 22:08:23.567830	128.119.245.12	10.12.189.164	HTTP	718	HTTP/1.1 200 OK (text/html)
483	2026-04-27 22:08:23.655452	10.12.189.164	128.119.245.12	HTTP	498	GET /favicon.ico HTTP/1.1

wireshark_Wi-Fi012DO3.pcapng

Packets: 1250 · Displayed: 3 (0.2%) · Dropped: 0 (0.0%) Profile: Default

Q12. How many HTTP GET request messages did your browser send? Which packet contains the GET for the Bill of Rights?

- 1 GET request, Packet 459

Q13. Which packet contains the status code and phrase associated with the response?

- The server's response with the status line is in **Packet 467**.

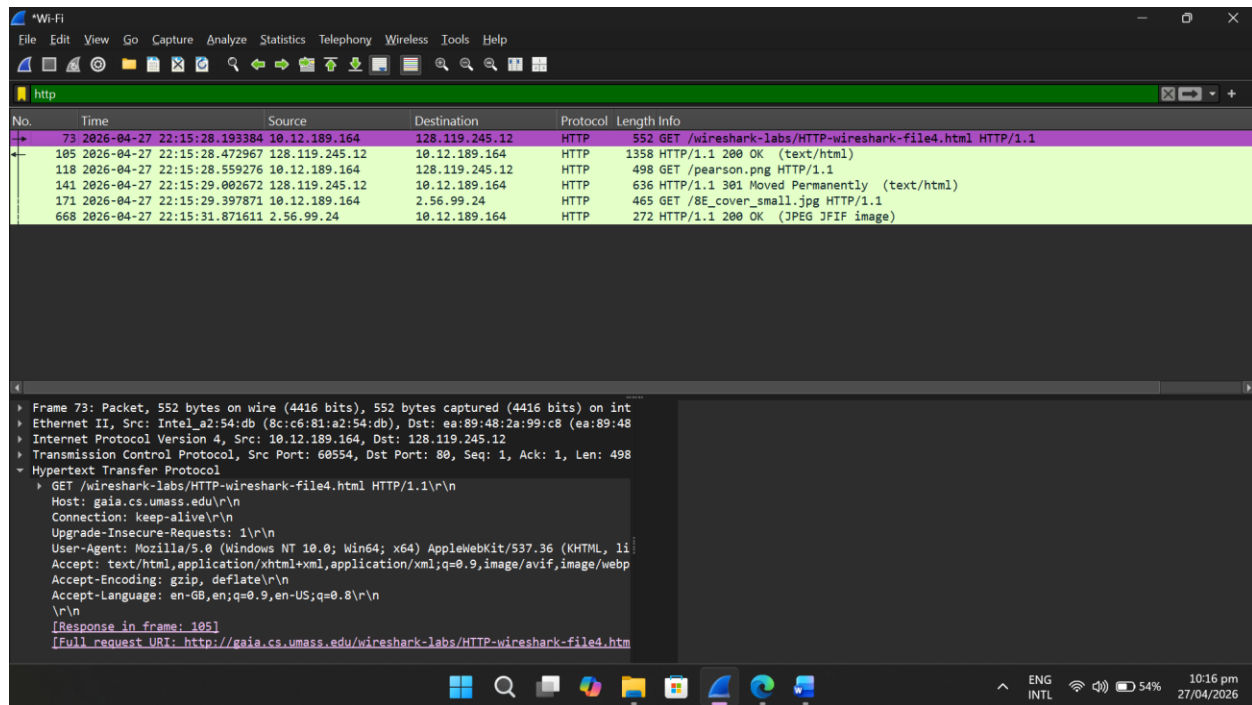
Q14. What is the status code and phrase in the response?

- Status code: **200**
- Phrase: **OK**
- Full line: HTTP/1.1 200 OK

Q15. How many data-containing TCP segments were needed to carry the single HTTP response and the text of the Bill of Rights?

- ~4–5 TCP segments (check packets after 467)

4. HTML Documents with Embedded Objects



Q16. How many HTTP GET request messages did your browser send? To which Internet addresses were they sent?

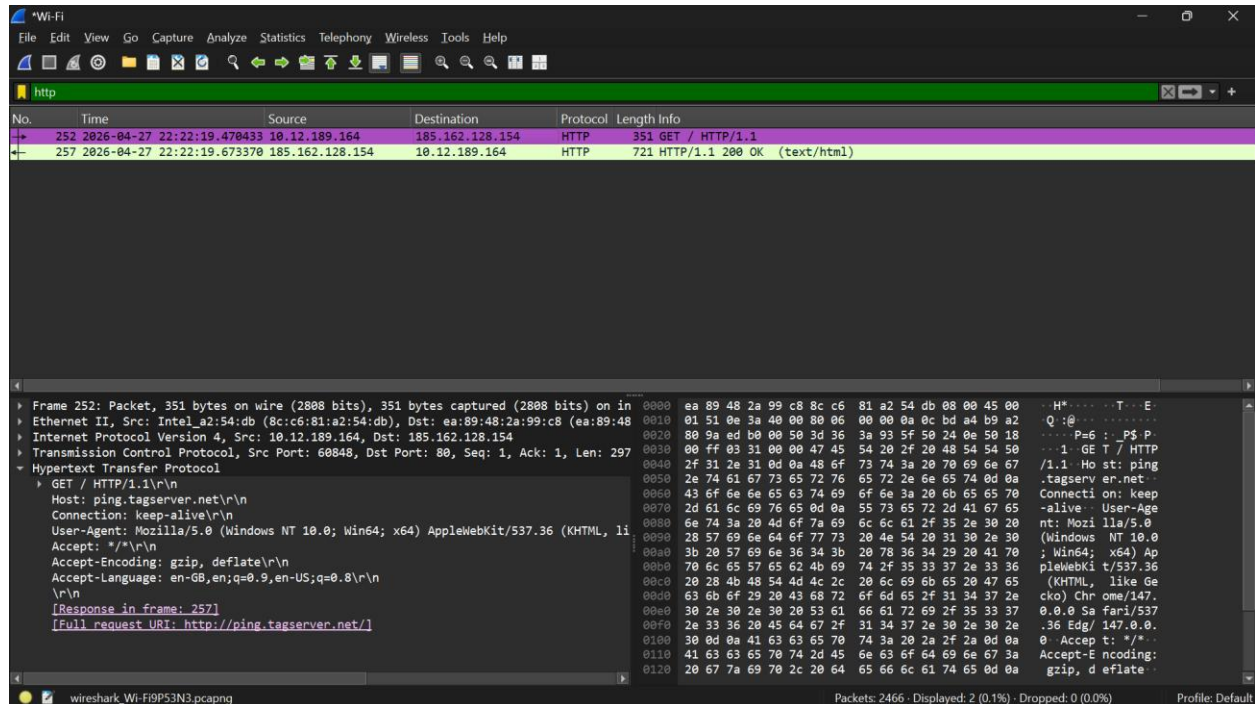
- browser sent **3 GET requests**:
 1. GET /wireshark-labs/HTTP-wireshark-file4.html → **128.119.245.12 (gaia.cs.umass.edu)**
 2. GET /pearson.png → **128.119.245.12 (same server)**
 3. GET /8E_cover_small.jpg → **French server (different IP address)**

So, one HTML file + two images = **3 GETs total**, across **two different servers**.

Q17. Were the two images downloaded serially or in parallel? Explain.

- Images downloaded in **parallel** (requests overlap in time)

5 HTTP Authentication



Q18. What is the server's response (status code and phrase) to the initial HTTP GET?

- TServer response = **401 Unauthorized**

Q19. What new field is included in the second HTTP GET message?

- The second GET request (after you enter username/password) includes an **Authorization** header:

Code

Authorization: Basic d2lyZXNoYXJrLXN0dWRlbnRzOm5ldHdvcm5=